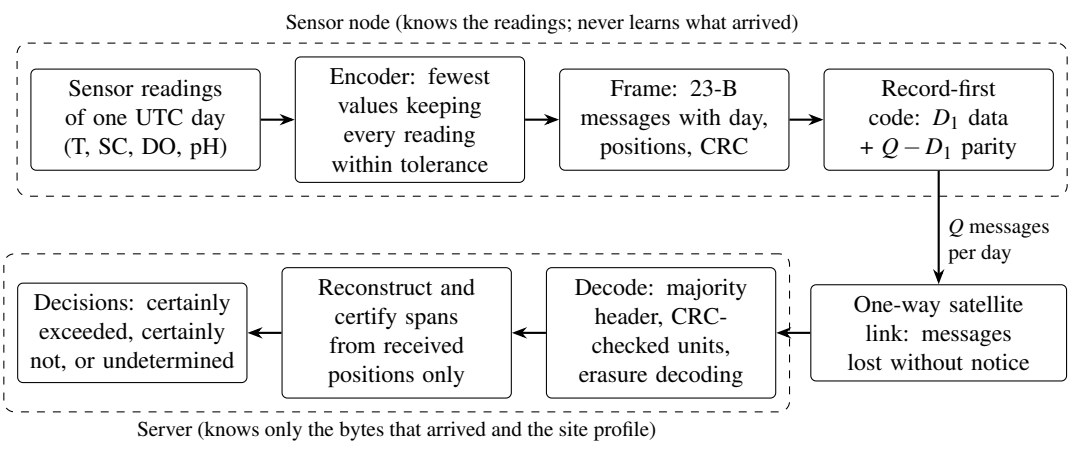
Sensor node (knows the readings; never learns what arrived)
Sensor readings of one UTC day (T, SC, DO, pH)
Encoder: fewest values keeping every reading within tolerance
Frame: 23-B messages with day, positions, CRC
Record-first code: D_1 data + $Q - D_1$ parity
Q messages per day
One-way satellite link: messages lost without notice
Decode: majority header, CRC-checked units, erasure decoding
Reconstruct and certify spans from received positions only
Decisions: certainly exceeded, certainly not, or undetermined
Server (knows only the bytes that arrived and the site profile)